

HIL Testbed — Consolidated Diagnostics Report

RPi/HIL directory: D:\My Files\Personal Projects\HIL-Testbed\outputs
(RPi)\S2-HIL-Arduino\c53b876e-9d46-4e7f-8686-76f3be9aefb6\publisher
Laptop/dry-run directory: D:\My Files\Personal
Projects\HIL-Testbed\outputs\publisher\1-MV-rural--2-sw_S2_Laptop_Dry

Script	Status
Timing validation + analysis	OK
Outlier detection	OK
Scenario comparison	OK
Q_initial clip diagnostic (H1/H2)	OK
Coordination ground-truth check	OK

Timing validation + analysis

TIMING DATA VALIDATION + ANALYSIS

Loading RPi data from: D:\My Files\Personal Projects\HIL-Testbed\outputs (RPi)\S2-HIL-Arduino\c53b876e-9d46-4e7f-8686-7
Loading laptop data from: D:\My Files\Personal Projects\HIL-Testbed\outputs\publisher\1-MV-rural--2-sw_S2_Laptop_Dry\sc

VALIDATION CHECKS

4 issue(s) found:

- [RPi/volt_var_local] 30 values exceed 10x-IQR fence (251.99 ms) – max=3450.00 ms (heuristic flag, not necessarily a)
- [RPi/volt_var_coord] 2280 values exceed 10x-IQR fence (277.07 ms) – max=3977.24 ms (heuristic flag, not necessarily)
- [Laptop/volt_var_local] 48 values exceed 10x-IQR fence (346.18 ms) – max=3970.93 ms (heuristic flag, not necessarily)
- [Laptop/volt_var_coord] 40 values exceed 10x-IQR fence (644.05 ms) – max=52639.57 ms (heuristic flag, not necessarily)

ANALYSIS

--- A. Per-scenario t_total_ms distribution [RPi] (ms) ---

volt_var_local:

[Full distribution, all timesteps incl. t=0]
n=35136 mean=231.597 std=25.862 min=227.271 max=3449.998
median=231.092 p5=228.862 p25=230.105 p75=232.095 p95=233.484 p99=235.109
First timestep (t=0): 3449.998 ms (14.9x steady-state median)
[Steady-state distribution, excludes first timestep only]
n=35135 mean=231.505 std=19.340 min=227.271 max=1945.230
median=231.092 p5=228.862 p25=230.105 p75=232.095 p95=233.483 p99=235.107

volt_var_coord:

[Full distribution, all timesteps incl. t=0]
n=35136 mean=242.526 std=39.926 min=231.898 max=3977.235
median=238.411 p5=234.685 p25=237.065 p75=240.702 p95=278.165 p99=282.027
First timestep (t=0): 292.839 ms (1.2x steady-state median)
[Steady-state distribution, excludes first timestep only]
n=35135 mean=242.525 std=39.925 min=231.898 max=3977.235
median=238.411 p5=234.685 p25=237.065 p75=240.702 p95=278.163 p99=282.025

--- A. Per-scenario t_total_ms distribution [Laptop] (ms) ---

volt_var_local:

[Full distribution, all timesteps incl. t=0]
n=35136 mean=119.564 std=45.688 min=51.983 max=3970.927
median=117.215 p5=94.830 p25=106.859 p75=128.615 p95=144.701 p99=158.583
First timestep (t=0): 3970.927 ms (33.9x steady-state median)
[Steady-state distribution, excludes first timestep only]
n=35135 mean=119.455 std=40.807 min=51.983 max=2331.101
median=117.215 p5=94.830 p25=106.859 p75=128.614 p95=144.697 p99=158.517

volt_var_coord:

[Full distribution, all timesteps incl. t=0]
n=35136 mean=119.724 std=476.118 min=58.338 max=52639.566
median=89.487 p5=77.651 p25=83.365 p75=134.336 p95=173.201 p99=271.565
First timestep (t=0): 437.401 ms (4.9x steady-state median)
[Steady-state distribution, excludes first timestep only]
n=35135 mean=119.715 std=476.122 min=58.338 max=52639.566
median=89.486 p5=77.651 p25=83.365 p75=134.334 p95=173.180 p99=271.466

--- B. Scenario 4 decomposition: t_total_ms - hil_latency_ms [RPi] (ms) ---

volt_var_coord (non-serial cost = PF solves + Schur solve + dynamics):

n=35136 mean=88.437 std=39.905 min=78.753 max=3821.340
median=84.280 p5=81.344 p25=82.945 p75=86.246 p95=124.188 p99=127.301
share of t_total_ms that is serial latency: 63.5%

volt_var_local (non-serial cost = PF solves + Schur solve + dynamics):

n=35136 mean=77.533 std=25.840 min=75.194 max=3295.363
median=77.095 p5=75.746 p25=76.307 p75=77.663 p95=78.495 p99=80.709
share of t_total_ms that is serial latency: 66.5%

--- B. Scenario 4 decomposition: t_total_ms - hil_latency_ms [Laptop] (ms) ---

volt_var_coord (non-serial cost = PF solves + Schur solve + dynamics):

n=35136 mean=119.724 std=476.118 min=58.338 max=52639.566
median=89.487 p5=77.651 p25=83.365 p75=134.336 p95=173.201 p99=271.565
share of t_total_ms that is serial latency: 0.0%

volt_var_local (non-serial cost = PF solves + Schur solve + dynamics):

n=35136 mean=119.564 std=45.688 min=51.983 max=3970.927
median=117.215 p5=94.830 p25=106.859 p75=128.615 p95=144.701 p99=158.583
share of t_total_ms that is serial latency: 0.0%

--- C. HIL vs dry-run t_total_ms comparison (ms) ---

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baseline: missing in one environment, skipped
oltc: missing in one environment, skipped
svc: missing in one environment, skipped
volt_var_local:
  RPi : mean=231.597 ms median=231.092 ms
  Laptop : mean=119.564 ms median=117.215 ms
  RPi/Laptop ratio (mean): 1.94x
volt_var_coord:
  RPi : mean=242.526 ms median=238.411 ms
  Laptop : mean=119.724 ms median=89.487 ms
  RPi/Laptop ratio (mean): 2.03x

--- D. Correlation with control activity [RPi] ---
volt_var_coord - coordination_active / curtailment_needed:
  coordination_active=True : n=34954 mean=242.536 ms
  coordination_active=False : n=182 mean=240.721 ms
  curtailment_needed=True : n=27853 mean=243.102 ms
  curtailment_needed=False : n=7283 mean=240.322 ms
volt_var_local - coordination_active / curtailment_needed:
  coordination_active=True : n=0
  coordination_active=False : n=35136 mean=231.597 ms
  curtailment_needed=True : n=35136 mean=231.597 ms
  curtailment_needed=False : n=0

--- D. Correlation with control activity [Laptop] ---
volt_var_coord - coordination_active / curtailment_needed:
  coordination_active=True : n=34954 mean=119.547 ms
  coordination_active=False : n=182 mean=153.627 ms
  curtailment_needed=True : n=27851 mean=118.695 ms
  curtailment_needed=False : n=7285 mean=123.656 ms
volt_var_local - coordination_active / curtailment_needed:
  coordination_active=True : n=0
  coordination_active=False : n=35136 mean=119.564 ms
  curtailment_needed=True : n=35136 mean=119.564 ms
  curtailment_needed=False : n=0

--- E. Outlier scan + temporal trend check [RPi] ---
volt_var_local:
  Top 5 slowest timesteps:
    t=0 t_total_ms=3449.998, converged=True, coordination_active=False, curtailment_needed=True
    t=33658 t_total_ms=1945.230, converged=True, coordination_active=False, curtailment_needed=True
    t=30334 t_total_ms=1788.150, converged=True, coordination_active=False, curtailment_needed=True
    t=27324 t_total_ms=1644.090, converged=True, coordination_active=False, curtailment_needed=True
    t=23701 t_total_ms=1482.079, converged=True, coordination_active=False, curtailment_needed=True
  Temporal trend: slope=-0.000002 ms/timestep, corr(t, t_total_ms)=-0.0009
  Implied drift over this run: -0.085 ms (-0.0% relative to mean)
  No strong linear trend detected (|corr| <= 0.3) - variation looks more like noise/isolated spikes than systematic
  Quartile-block means (chronological): Q1=231.729, Q2=231.505, Q3=231.350, Q4=231.804
volt_var_coord:
  Top 5 slowest timesteps:
    t=33044 t_total_ms=3977.235, converged=True, coordination_active=True, curtailment_needed=True
    t=22909 t_total_ms=3512.096, converged=True, coordination_active=True, curtailment_needed=True
    t=11782 t_total_ms=2993.932, converged=True, coordination_active=True, curtailment_needed=False
    t=1274 t_total_ms=2790.545, converged=True, coordination_active=True, curtailment_needed=True
    t=6720 t_total_ms=2781.100, converged=True, coordination_active=True, curtailment_needed=True
  Temporal trend: slope=0.000035 ms/timestep, corr(t, t_total_ms)=0.0090
  Implied drift over this run: +1.243 ms (+0.5% relative to mean)
  No strong linear trend detected (|corr| <= 0.3) - variation looks more like noise/isolated spikes than systematic
  Quartile-block means (chronological): Q1=241.761, Q2=242.467, Q3=243.326, Q4=242.551

--- E. Outlier scan + temporal trend check [Laptop] ---
volt_var_local:
  Top 5 slowest timesteps:
    t=0 t_total_ms=3970.927, converged=True, coordination_active=False, curtailment_needed=True
    t=34241 t_total_ms=2331.101, converged=True, coordination_active=False, curtailment_needed=True
    t=31691 t_total_ms=2198.325, converged=True, coordination_active=False, curtailment_needed=True
    t=32943 t_total_ms=2193.820, converged=True, coordination_active=False, curtailment_needed=True
    t=29012 t_total_ms=2167.546, converged=True, coordination_active=False, curtailment_needed=True
  Temporal trend: slope=0.000402 ms/timestep, corr(t, t_total_ms)=0.0893
  Implied drift over this run: +14.137 ms (+11.8% relative to mean)
  No strong linear trend detected (|corr| <= 0.3) - variation looks more like noise/isolated spikes than systematic
  Quartile-block means (chronological): Q1=121.210, Q2=112.372, Q3=113.440, Q4=131.236
volt_var_coord:
  Top 5 slowest timesteps:
    t=3519 t_total_ms=52639.566, converged=True, coordination_active=True, curtailment_needed=True
    t=3518 t_total_ms=36092.052, converged=True, coordination_active=True, curtailment_needed=True
    t=3516 t_total_ms=28625.403, converged=True, coordination_active=True, curtailment_needed=True
    t=3517 t_total_ms=25068.138, converged=True, coordination_active=True, curtailment_needed=True
    t=3520 t_total_ms=24344.920, converged=True, coordination_active=True, curtailment_needed=True
  Temporal trend: slope=-0.003784 ms/timestep, corr(t, t_total_ms)=-0.0806
  Implied drift over this run: -132.948 ms (-111.0% relative to mean)

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No strong linear trend detected ($|corr| \leq 0.3$) – variation looks more like noise/isolated spikes than systematic
Quartile-block means (chronological): Q1=188.345, Q2=114.925, Q3=85.670, Q4=89.957

--- F. Run coverage [RPI] ---

This run covers 35136 timesteps (full year (366 days), 15-min resolution). All seasons are represented. Distribution

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END OF REPORT
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[WARN] baseline: no file found in D:\My Files\Personal Projects\HIL-Testbed\outputs (RPI)\S2-HIL-Arduino\c53b876e-9d46-4e7a-8b3a-1b2b2b2b2b2b
[WARN] oltc: no file found in D:\My Files\Personal Projects\HIL-Testbed\outputs (RPI)\S2-HIL-Arduino\c53b876e-9d46-4e7a-8b3a-1b2b2b2b2b2b
[WARN] svc: no file found in D:\My Files\Personal Projects\HIL-Testbed\outputs (RPI)\S2-HIL-Arduino\c53b876e-9d46-4e7a-8b3a-1b2b2b2b2b2b
[WARN] baseline: no file found in D:\My Files\Personal Projects\HIL-Testbed\outputs\publisher\1-MV-rural--2-sw_S2_Laptop_D
[WARN] oltc: no file found in D:\My Files\Personal Projects\HIL-Testbed\outputs\publisher\1-MV-rural--2-sw_S2_Laptop_D
[WARN] svc: no file found in D:\My Files\Personal Projects\HIL-Testbed\outputs\publisher\1-MV-rural--2-sw_S2_Laptop_D

Outlier detection

OUTLIER DIAGNOSTIC FOLLOW-UP

Loading RPi data from: D:\My Files\Personal Projects\HIL-Testbed\outputs (RPi)\S2-HIL-Arduino\c53b876e-9d46-4e7f-8686-7
Loading laptop data from: D:\My Files\Personal Projects\HIL-Testbed\outputs\publisher\1-MV-rural--2-sw_S2_Laptop_Dry\sc

ENVIRONMENT: RPi

--- volt_var_local: 29 outlier(s) (t=0 excluded) ---

Top 20 outliers:

t=33658, t_total_ms=1945.2303990001383, timestamp=2016-12-16T14:30:00+01:00, converged=True, tap_changed=None, tap_a
t=30334, t_total_ms=1788.1504440010758, timestamp=2016-11-11T23:30:00+01:00, converged=True, tap_changed=None, tap_a
t=27324, t_total_ms=1644.0895489977265, timestamp=2016-10-11T16:00:00+02:00, converged=True, tap_changed=None, tap_a
t=23701, t_total_ms=1482.0794229999592, timestamp=2016-09-03T22:15:00+02:00, converged=True, tap_changed=None, tap_a
t=22053, t_total_ms=1397.3241490002692, timestamp=2016-08-17T18:15:00+02:00, converged=True, tap_changed=None, tap_a
t=15780, t_total_ms=1113.970631000484, timestamp=2016-06-13T10:00:00+02:00, converged=True, tap_changed=None, tap_a
t=14059, t_total_ms=1030.551101999663, timestamp=2016-05-26T11:45:00+02:00, converged=True, tap_changed=None, tap_a
t=10578, t_total_ms=867.1659139999974, timestamp=2016-04-20T05:30:00+02:00, converged=True, tap_changed=None, tap_a
t=8918, t_total_ms=792.8493160006838, timestamp=2016-04-02T22:30:00+02:00, converged=True, tap_changed=None, tap_at
t=7082, t_total_ms=714.8276249999981, timestamp=2016-03-14T18:30:00+01:00, converged=True, tap_changed=None, tap_at
t=6111, t_total_ms=662.193629000285, timestamp=2016-03-04T15:45:00+01:00, converged=True, tap_changed=None, tap_att
t=2588, t_total_ms=504.90756699991834, timestamp=2016-01-27T23:00:00+01:00, converged=True, tap_changed=None, tap_a
t=2390, t_total_ms=497.65402499997435, timestamp=2016-01-25T21:30:00+01:00, converged=True, tap_changed=None, tap_a
t=923, t_total_ms=423.32038299991837, timestamp=2016-01-10T14:45:00+01:00, converged=True, tap_changed=None, tap_at
t=376, t_total_ms=396.0021370000959, timestamp=2016-01-04T22:00:00+01:00, converged=True, tap_changed=None, tap_att
t=29324, t_total_ms=261.2057559999812, timestamp=2016-11-01T11:00:00+01:00, converged=True, tap_changed=None, tap_a
t=30930, t_total_ms=259.8968940001214, timestamp=2016-11-18T04:30:00+01:00, converged=True, tap_changed=None, tap_a
t=4350, t_total_ms=254.00053200019101, timestamp=2016-02-15T07:30:00+01:00, converged=True, tap_changed=None, tap_a
t=32217, t_total_ms=253.4231550016557, timestamp=2016-12-01T14:15:00+01:00, converged=True, tap_changed=None, tap_a
t=25234, t_total_ms=253.32240799980354, timestamp=2016-09-19T21:30:00+02:00, converged=True, tap_changed=None, tap_a

[RPi/volt_var_local] Periodicity check:

n=29 outliers, t range [376, 33658]

Gaps between consecutive outlier timesteps: mean=1188.6 std=662.2 min=60 max=2462

Coefficient of variation of gaps: 0.557

OBSERVATION: gaps are irregular (CV >= 0.4) – no strong evidence of periodicity; outliers look more scattered/random

[RPi/volt_var_local] Calendar placement of outliers:

Outlier count by month:

Month 1: 6
Month 2: 1
Month 3: 3
Month 4: 2
Month 5: 3
Month 6: 1
Month 7: 2
Month 8: 1
Month 9: 4
Month 10: 1
Month 11: 3
Month 12: 2

--- volt_var_coord: 2279 outlier(s) (t=0 excluded) ---

Top 20 outliers:

t=33044, t_total_ms=3977.2353759981343, timestamp=2016-12-10T05:00:00+01:00, converged=True, tap_changed=None, tap_a
t=22909, t_total_ms=3512.0961670036195, timestamp=2016-08-26T16:15:00+02:00, converged=True, tap_changed=None, tap_a
t=11782, t_total_ms=2993.931838998833, timestamp=2016-05-02T18:30:00+02:00, converged=True, tap_changed=None, tap_a
t=1274, t_total_ms=2790.5445210017206, timestamp=2016-01-14T06:30:00+01:00, converged=True, tap_changed=None, tap_a
t=6720, t_total_ms=2781.099781001103, timestamp=2016-03-11T00:00:00+01:00, converged=True, tap_changed=None, tap_at
t=4838, t_total_ms=2692.988410999533, timestamp=2016-02-20T09:30:00+01:00, converged=True, tap_changed=None, tap_at
t=1039, t_total_ms=304.7115329973167, timestamp=2016-01-11T19:45:00+01:00, converged=True, tap_changed=None, tap_at
t=1116, t_total_ms=303.4038910009258, timestamp=2016-01-12T15:00:00+01:00, converged=True, tap_changed=None, tap_at
t=4512, t_total_ms=301.41439200087916, timestamp=2016-02-17T00:00:00+01:00, converged=True, tap_changed=None, tap_a
t=31766, t_total_ms=300.6428529988625, timestamp=2016-11-26T21:30:00+01:00, converged=True, tap_changed=None, tap_a
t=23336, t_total_ms=300.42065299494425, timestamp=2016-08-31T03:00:00+02:00, converged=True, tap_changed=None, tap_a
t=3732, t_total_ms=300.0568249990465, timestamp=2016-02-08T21:00:00+01:00, converged=True, tap_changed=None, tap_at
t=3846, t_total_ms=298.38685099821305, timestamp=2016-02-10T01:30:00+01:00, converged=True, tap_changed=None, tap_a
t=4019, t_total_ms=298.310150999896, timestamp=2016-02-11T20:45:00+01:00, converged=True, tap_changed=None, tap_att
t=770, t_total_ms=297.96488099964336, timestamp=2016-01-09T00:30:00+01:00, converged=True, tap_changed=None, tap_at
t=8358, t_total_ms=297.43441000027815, timestamp=2016-03-28T02:30:00+02:00, converged=True, tap_changed=None, tap_a
t=32311, t_total_ms=297.383972996613, timestamp=2016-12-02T13:45:00+01:00, converged=True, tap_changed=None, tap_at
t=33872, t_total_ms=297.3708479985362, timestamp=2016-12-18T20:00:00+01:00, converged=True, tap_changed=None, tap_a
t=4959, t_total_ms=297.2314299986465, timestamp=2016-02-21T15:45:00+01:00, converged=True, tap_changed=None, tap_at
t=29760, t_total_ms=297.15370800113305, timestamp=2016-11-06T00:00:00+01:00, converged=True, tap_changed=None, tap_a

[RPi/volt_var_coord] Periodicity check:

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n=2279 outliers, t range [5, 35083]
Gaps between consecutive outlier timestamps: mean=15.4 std=18.7 min=1 max=151
Coefficient of variation of gaps: 1.216
OBSERVATION: gaps are irregular (CV >= 0.4) – no strong evidence of periodicity; outliers look more scattered/random

[RPi/volt_var_coord] Calendar placement of outliers:
  Outlier count by month:
    Month 1: 143
    Month 2: 107
    Month 3: 136
    Month 4: 218
    Month 5: 183
    Month 6: 189
    Month 7: 254
    Month 8: 235
    Month 9: 239
    Month 10: 181
    Month 11: 189
    Month 12: 205

[RPi] Cross-scenario outlier clustering (within +/-5 timesteps):
  volt_var_local vs volt_var_coord: 16/2279 of volt_var_coord's outliers (0.7%) fall within 5 steps of an volt_var_local
  Example matches: t=2327~2330, t=30336~30334, t=14059~14059, t=23884~23886, t=30333~30334

[RPi] Exact-zero t_total_ms anomaly check:
  None found in Scenario 4 files.

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ENVIRONMENT: Laptop
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--- volt_var_local: 47 outlier(s) (t=0 excluded) ---
Top 20 outliers:
  t=34241, t_total_ms=2331.1013000202365, timestamp=2016-12-22T16:15:00+01:00, converged=True, tap_changed=None, tap_a
  t=31691, t_total_ms=2198.325300007127, timestamp=2016-11-26T02:45:00+01:00, converged=True, tap_changed=None, tap_a
  t=32943, t_total_ms=2193.8197999843396, timestamp=2016-12-09T03:45:00+01:00, converged=True, tap_changed=None, tap_a
  t=29012, t_total_ms=2167.5463000428863, timestamp=2016-10-29T06:00:00+02:00, converged=True, tap_changed=None, tap_a
  t=27888, t_total_ms=1981.077299977187, timestamp=2016-10-17T13:00:00+02:00, converged=True, tap_changed=None, tap_a
  t=30475, t_total_ms=1910.9447000082582, timestamp=2016-11-13T10:45:00+01:00, converged=True, tap_changed=None, tap_a
  t=25236, t_total_ms=1724.1811999701895, timestamp=2016-09-19T22:00:00+02:00, converged=True, tap_changed=None, tap_a
  t=26534, t_total_ms=1522.2585000446998, timestamp=2016-10-03T10:30:00+02:00, converged=True, tap_changed=None, tap_a
  t=26268, t_total_ms=1520.3765999758616, timestamp=2016-09-30T16:00:00+02:00, converged=True, tap_changed=None, tap_a
  t=23033, t_total_ms=1464.5191000308841, timestamp=2016-08-27T23:15:00+02:00, converged=True, tap_changed=None, tap_a
  t=23990, t_total_ms=1435.3700000210665, timestamp=2016-09-06T22:30:00+02:00, converged=True, tap_changed=None, tap_a
  t=20990, t_total_ms=1378.4937000018544, timestamp=2016-08-06T16:30:00+02:00, converged=True, tap_changed=None, tap_a
  t=19730, t_total_ms=1287.5822999631055, timestamp=2016-07-24T13:30:00+02:00, converged=True, tap_changed=None, tap_a
  t=18720, t_total_ms=1219.3110999651253, timestamp=2016-07-14T01:00:00+02:00, converged=True, tap_changed=None, tap_a
  t=16468, t_total_ms=1134.0848000254482, timestamp=2016-06-20T14:00:00+02:00, converged=True, tap_changed=None, tap_a
  t=15597, t_total_ms=1070.3904000110924, timestamp=2016-06-11T12:15:00+02:00, converged=True, tap_changed=None, tap_a
  t=14759, t_total_ms=1053.6980999750085, timestamp=2016-06-02T18:45:00+02:00, converged=True, tap_changed=None, tap_a
  t=13341, t_total_ms=959.4749999814667, timestamp=2016-05-19T00:15:00+02:00, converged=True, tap_changed=None, tap_a
  t=12306, t_total_ms=929.6125000109896, timestamp=2016-05-08T05:30:00+02:00, converged=True, tap_changed=None, tap_a
  t=11474, t_total_ms=900.6003999966197, timestamp=2016-04-29T13:30:00+02:00, converged=True, tap_changed=None, tap_a

[Laptop/volt_var_local] Periodicity check:
  n=47 outliers, t range [2, 34241]
  Gaps between consecutive outlier timestamps: mean=744.3 std=522.0 min=2 max=2252
  Coefficient of variation of gaps: 0.701
  OBSERVATION: gaps are irregular (CV >= 0.4) – no strong evidence of periodicity; outliers look more scattered/random

[Laptop/volt_var_local] Calendar placement of outliers:
  Outlier count by month:
    Month 1: 10
    Month 2: 6
    Month 3: 7
    Month 4: 4
    Month 5: 2
    Month 6: 3
    Month 7: 2
    Month 8: 2
    Month 9: 3
    Month 10: 3
    Month 11: 2
    Month 12: 3

--- volt_var_coord: 40 outlier(s) (t=0 excluded) ---
Top 20 outliers:
  t=3519, t_total_ms=52639.56640003016, timestamp=2016-02-06T15:45:00+01:00, converged=True, tap_changed=None, tap_at
  t=3518, t_total_ms=36092.051599989645, timestamp=2016-02-06T15:30:00+01:00, converged=True, tap_changed=None, tap_a
  t=3516, t_total_ms=28625.402799982112, timestamp=2016-02-06T15:00:00+01:00, converged=True, tap_changed=None, tap_a
  t=3517, t_total_ms=25068.13760002842, timestamp=2016-02-06T15:15:00+01:00, converged=True, tap_changed=None, tap_at

```

t=3520, t_total_ms=24344.92030000547, timestamp=2016-02-06T16:00:00+01:00, converged=True, tap_changed=None, tap_at
t=3515, t_total_ms=17936.91020004917, timestamp=2016-02-06T14:45:00+01:00, converged=True, tap_changed=None, tap_at
t=3523, t_total_ms=14684.838700050022, timestamp=2016-02-06T16:45:00+01:00, converged=True, tap_changed=None, tap_a
t=3513, t_total_ms=14012.010899954475, timestamp=2016-02-06T14:15:00+01:00, converged=True, tap_changed=None, tap_a
t=3521, t_total_ms=12801.64750001859, timestamp=2016-02-06T16:15:00+01:00, converged=True, tap_changed=None, tap_at
t=3803, t_total_ms=12233.651600021403, timestamp=2016-02-09T14:45:00+01:00, converged=True, tap_changed=None, tap_a
t=3522, t_total_ms=11330.644000030588, timestamp=2016-02-06T16:30:00+01:00, converged=True, tap_changed=None, tap_a
t=3510, t_total_ms=9659.594999975525, timestamp=2016-02-06T13:30:00+01:00, converged=True, tap_changed=None, tap_at
t=3801, t_total_ms=9196.074200037401, timestamp=2016-02-09T14:15:00+01:00, converged=True, tap_changed=None, tap_at
t=3524, t_total_ms=8960.524900001474, timestamp=2016-02-06T17:00:00+01:00, converged=True, tap_changed=None, tap_at
t=3804, t_total_ms=8722.157999989577, timestamp=2016-02-09T15:00:00+01:00, converged=True, tap_changed=None, tap_at
t=3514, t_total_ms=8636.156900029164, timestamp=2016-02-06T14:30:00+01:00, converged=True, tap_changed=None, tap_at
t=3802, t_total_ms=7797.434000007343, timestamp=2016-02-09T14:30:00+01:00, converged=True, tap_changed=None, tap_at
t=3800, t_total_ms=7055.790200014599, timestamp=2016-02-09T14:00:00+01:00, converged=True, tap_changed=None, tap_at
t=34918, t_total_ms=4382.600100012496, timestamp=2016-12-29T17:30:00+01:00, converged=True, tap_changed=None, tap_a
t=31195, t_total_ms=4312.400599999819, timestamp=2016-11-20T22:45:00+01:00, converged=True, tap_changed=None, tap_a

[Laptop/volt_var_coord] Periodicity check:

n=40 outliers, t range [313, 34918]

Gaps between consecutive outlier timesteps: mean=887.3 std=1619.1 min=1 max=7284

Coefficient of variation of gaps: 1.825

OBSERVATION: gaps are irregular (CV >= 0.4) – no strong evidence of periodicity; outliers look more scattered/random

[Laptop/volt_var_coord] Calendar placement of outliers:

Outlier count by month:

Month 1: 3
Month 2: 24
Month 3: 1
Month 4: 1
Month 5: 3
Month 7: 4
Month 10: 2
Month 11: 1
Month 12: 1

OBSERVATION: month 2 alone accounts for 60% of outliers – possible seasonal concentration.

[Laptop] Cross-scenario outlier clustering (within +/-5 timesteps):

volt_var_local vs volt_var_coord: 9/40 of volt_var_coord's outliers (22.5%) fall within 5 steps of an volt_var_local

Example matches: t=3519-3514, t=3518-3514, t=3516-3514, t=3517-3514, t=3515-3514

[Laptop] Exact-zero t_total_ms anomaly check:

None found in Scenario 4 files.

=====
END OF REPORT
=====

[WARN] baseline: no file found in D:\My Files\Personal Projects\HIL-Testbed\outputs (RPI)\S2-HIL-Arduino\c53b876e-9d46-4e7b-8000-000000000000

[WARN] oltc: no file found in D:\My Files\Personal Projects\HIL-Testbed\outputs (RPI)\S2-HIL-Arduino\c53b876e-9d46-4e7b-8000-000000000000

[WARN] svc: no file found in D:\My Files\Personal Projects\HIL-Testbed\outputs (RPI)\S2-HIL-Arduino\c53b876e-9d46-4e7b-8000-000000000000

[WARN] baseline: no file found in D:\My Files\Personal Projects\HIL-Testbed\outputs\publisher\1-MV-rural--2-sw_S2_Laptop_D

[WARN] oltc: no file found in D:\My Files\Personal Projects\HIL-Testbed\outputs\publisher\1-MV-rural--2-sw_S2_Laptop_D

[WARN] svc: no file found in D:\My Files\Personal Projects\HIL-Testbed\outputs\publisher\1-MV-rural--2-sw_S2_Laptop_D

Scenario comparison

```
#####
# ENVIRONMENT: RPi - D:\My Files\Personal Projects\HIL-Testbed\outputs (RPi)\S2-HIL-Arduino\c53b876e-9d46-4e7f-8686-7
#####
[WARN] baseline: not found in D:\My Files\Personal Projects\HIL-Testbed\outputs (RPi)\S2-HIL-Arduino\c53b876e-9d46-4e7f-8686-7
[WARN] oltc: not found in D:\My Files\Personal Projects\HIL-Testbed\outputs (RPi)\S2-HIL-Arduino\c53b876e-9d46-4e7f-8686-7
[WARN] svc: not found in D:\My Files\Personal Projects\HIL-Testbed\outputs (RPi)\S2-HIL-Arduino\c53b876e-9d46-4e7f-8686-7

=====
C. Extreme State Timeseries [RPi]
=====

--- C1. max_vm_pu distribution across scenarios ---
Scenario          mean    median    p95      p99      max    > 1.05 (%)    > 1.08 (%)
Volt-Var Local (4A)  1.0574  1.0574  1.0617  1.0621  1.0625    99.2%        0.0%
Volt-Var Coord (4B)  1.0469  1.0506  1.0574  1.0603  1.0644    52.6%        0.0%

--- C2. max_line_loading_pct distribution across scenarios ---
Scenario          mean    median    p95      p99      max    > 100% (steps)
Volt-Var Local (4A)  33.65   32.93   38.52   38.93   39.90         0
Volt-Var Coord (4B)  46.48   44.10   73.23   82.37   84.99         0

--- C3. Voltage envelope: how often does max_vm_pu exceed thresholds? ---
Scenario          >= 1.02    >= 1.05    >= 1.08    >= 1.10
Volt-Var Local (4A)  100.0%    99.2%     0.0%     0.0%
Volt-Var Coord (4B)  100.0%    52.6%     0.0%     0.0%

=====
D. Reactive Power Distribution - Scenario 4 [RPi]
=====

--- D1. Volt-Var Local (4A) - per-DER Q statistics ---
98 DERs, 35136 timesteps with Q data
DER    mean Q    std Q    min Q    max Q    % at limit (>0.95*max)
72     -0.0040    0.0005   -0.0049   -0.0018    14.7%
29     -0.0044    0.0006   -0.0053   -0.0017    14.1%
26     -0.0043    0.0006   -0.0052   -0.0017    13.1%
75     -0.0045    0.0006   -0.0054   -0.0018    13.0%
13     -0.0042    0.0005   -0.0051   -0.0019    12.5%
12     -0.0045    0.0006   -0.0054   -0.0020    12.2%
74     -0.0047    0.0006   -0.0056   -0.0019    12.1%
28     -0.0044    0.0006   -0.0053   -0.0017    12.0%
0      -0.0335    0.0043   -0.0410   -0.0153    11.9%
73     -0.0042    0.0005   -0.0051   -0.0018    11.3%

Overall Q saturation: mean q_saturated_count = 0.0 of 98 DERs per timestep
Total reactive energy (from summary): 15664.300575000101 MVar.h

--- D2. Volt-Var Local (4A) - Q budget utilisation over time ---
Total |Q| per timestep: mean=1.783 MVar std=0.293 max=2.727 min=0.792
Monthly mean total |Q| (MVar):
Month 1: 1.611
Month 2: 1.643
Month 3: 1.736
Month 4: 1.843
Month 5: 1.881
Month 6: 1.770
Month 7: 1.869
Month 8: 1.842
Month 9: 1.831
Month 10: 1.813
Month 11: 1.656
Month 12: 1.895

--- D1. Volt-Var Coord (4B) - per-DER Q statistics ---
98 DERs, 35136 timesteps with Q data
DER    mean Q    std Q    min Q    max Q    % at limit (>0.95*max)
42     -0.1955    0.0130   -0.1966   -0.0226    99.0%
41     -0.1954    0.0134   -0.1966    0.0419    98.9%
40     -0.1923    0.0340   -0.1966    0.1966    98.2%
61     -0.0703    0.0119   -0.0723    0.0532    96.8%
60     -0.0703    0.0119   -0.0723    0.0532    96.8%
56     -0.0703    0.0119   -0.0723    0.0533    96.8%
55     -0.0702    0.0120   -0.0723    0.0723    96.8%
51     -0.0687    0.0184   -0.0723    0.0723    96.0%
50     -0.0682    0.0201   -0.0723    0.0723    95.9%
48     -0.0666    0.0245   -0.0723    0.0723    95.3%

Overall Q saturation: mean q_saturated_count = 33.6 of 98 DERs per timestep
```

Total reactive energy (from summary): 72924.05547439973 MVar·h

--- D2. Volt-Var Coord (4B) – Q budget utilisation over time ---

Total |Q| per timestep: mean=8.302 MVar std=2.546 max=16.291 min=0.792

Monthly mean total |Q| (MVar):

Month 1: 8.050
Month 2: 8.091
Month 3: 8.186
Month 4: 8.319
Month 5: 8.696
Month 6: 7.705
Month 7: 8.165
Month 8: 8.015
Month 9: 8.187
Month 10: 8.410
Month 11: 7.638
Month 12: 10.105

=====

E. Active Power Curtailment – Scenario 4 [RPi]

=====

--- E1. Volt-Var Local (4A) – curtailment overview ---

Curtailment steps: 35136

Curtailed energy: 65138.2979 MWh

Curtailment as % of total DER generation: 100.000%

Per-DER curtailment breakdown (top 10 most curtailed):

All timesteps are curtailment steps – cannot derive uncurtailed reference from non-curtailment steps.

Curtailment steps by month:

Month 1: 2976 steps
Month 2: 2784 steps
Month 3: 2976 steps
Month 4: 2880 steps
Month 5: 2976 steps
Month 6: 2880 steps
Month 7: 2976 steps
Month 8: 2976 steps
Month 9: 2880 steps
Month 10: 2976 steps
Month 11: 2880 steps
Month 12: 2976 steps

--- E1. Volt-Var Coord (4B) – curtailment overview ---

Curtailment steps: 27853

Curtailed energy: 52961.5398 MWh

Curtailment as % of total DER generation: 81.306%

Per-DER curtailment breakdown (top 10 most curtailed):

DER	natural P (MW)	mean P on curt steps	curtailed MWh	curt steps
93	1.5003	0.1089	10054.9148	26933
91	1.2248	0.0889	8208.0937	26933
97	0.6124	0.0445	4104.0468	26933
0	0.4860	0.0366	3258.9630	27021
96	0.4374	0.0329	2933.0667	27021
92	0.4131	0.0311	2770.1185	27021
1	0.0000	0.0003	0.0000	0
2	0.0000	0.0003	0.0000	0
3	0.0000	0.0003	0.0000	0
4	0.0000	0.0003	0.0000	0

Sum of per-DER curtailed energy (curtailment steps only): 31329.2034 MWh

Summary curtailed_energy_mwh: 52961.5398 MWh (includes PT1-dynamics P lag on all steps, not just curtailment steps –

Curtailment steps by month:

Month 1: 1683 steps
Month 2: 1735 steps
Month 3: 2275 steps
Month 4: 2649 steps
Month 5: 2597 steps
Month 6: 2701 steps
Month 7: 2804 steps
Month 8: 2756 steps
Month 9: 2453 steps
Month 10: 2311 steps
Month 11: 2037 steps
Month 12: 1852 steps

=====

F. Export / Import Balance [RPi]

=====

=====

Note: grid_import_mw < 0 means the network is NET EXPORTING (DER generation exceeds local load + losses).

Scenario	Export steps	Import steps	Export %	Net export MWh	Net import MWh
Volt-Var Local (4A)	35136	0	100.0%	95785.50	0.00
Volt-Var Coord (4B)	35136	0	100.0%	107057.43	0.00

--- F2. Monthly net export comparison: baseline vs Scenario 4 ---

Month	Volt-Var Local (4A)	Volt-Var Coord (4B)
1	7447.36	9162.29
2	6967.51	8312.61
3	7832.55	8662.45
4	8111.15	8424.23
5	8450.08	8926.69
6	8226.80	8369.67
7	8554.66	8822.66
8	8591.66	8827.70
9	8081.94	8672.99
10	8300.63	9429.10
11	7653.74	8637.63
12	7567.42	10809.41

(values in MWh exported per month)

=====

G. Violation Frequency Map [RPi]

=====

--- G. Volt-Var Local (4A) ---

Total overvoltage bus-steps: 875048 | Total undervoltage bus-steps: 0

Top 10 most frequently overvoltage buses:

Bus	Count	% of all OV steps
44	35136	4.0%
45	35136	4.0%
46	35136	4.0%
47	35136	4.0%
43	35133	4.0%
42	35127	4.0%
68	33906	3.9%
67	33884	3.9%
66	33857	3.9%
98	33827	3.9%

No undervoltage bus-steps recorded.

--- G. Volt-Var Coord (4B) ---

Total overvoltage bus-steps: 478076 | Total undervoltage bus-steps: 0

Top 10 most frequently overvoltage buses:

Bus	Count	% of all OV steps
47	26633	5.6%
46	26612	5.6%
45	26513	5.5%
44	25921	5.4%
43	23250	4.9%
42	22336	4.7%
68	16645	3.5%
67	16618	3.5%
66	16577	3.5%
98	16509	3.5%

No undervoltage bus-steps recorded.

=====

H. Summary Comparison Table [RPi]

=====

Metric	Volt-Var Local (4A)	Volt-Var Coord (4B)
Violation steps	35136	26633
OV bus-steps	875048	478076
UV bus-steps	0	0
VDI	114277.7065	89459.9505
Total losses (MWh)	1850.4284	2755.2626
Reactive energy (MVar·h)	15664.3006	72924.0555
Curtailed energy (MWh)	65138.2979	52961.5398
DER generation (MWh)	65138.2979	65138.2979
Grid export (MWh)	95785.5019	107057.4259
Peak max_vm_pu	1.0625	1.0644
Worst min_vm_pu	1.0135	0.9826
Peak line loading (%)	39.9028	84.9948
Coordination rate	0.0000	0.9948
Q saturation rate	0.0000	0.9948
Elapsed (s)	21701.9570	19074.5835

--- H2. Loss penalty and reactive cost vs baseline ---

```
#####
# ENVIRONMENT: Laptop - D:\My Files\Personal Projects\HIL-Testbed\outputs\publisher\1-MV-rural--2-sw_S2_Laptop_Dry\sc
#####
[WARN] baseline: not found in D:\My Files\Personal Projects\HIL-Testbed\outputs\publisher\1-MV-rural--2-sw_S2_Laptop_
[WARN] oltc: not found in D:\My Files\Personal Projects\HIL-Testbed\outputs\publisher\1-MV-rural--2-sw_S2_Laptop_Dry\
[WARN] svc: not found in D:\My Files\Personal Projects\HIL-Testbed\outputs\publisher\1-MV-rural--2-sw_S2_Laptop_Dry\s
```

C. Extreme State Timeseries [Laptop]

--- C1. max_vm_pu distribution across scenarios ---

Scenario	mean	median	p95	p99	max	> 1.05 (%)	> 1.08 (%)
Volt-Var Local (4A)	1.0574	1.0574	1.0617	1.0621	1.0625	99.2%	0.0%
Volt-Var Coord (4B)	1.0469	1.0506	1.0574	1.0603	1.0644	52.6%	0.0%

--- C2. max_line_loading_pct distribution across scenarios ---

Scenario	mean	median	p95	p99	max	> 100% (steps)
Volt-Var Local (4A)	33.65	32.93	38.52	38.93	39.90	0
Volt-Var Coord (4B)	46.48	44.10	73.23	82.37	84.99	0

--- C3. Voltage envelope: how often does max_vm_pu exceed thresholds? ---

Scenario	>= 1.02	>= 1.05	>= 1.08	>= 1.10
Volt-Var Local (4A)	100.0%	99.2%	0.0%	0.0%
Volt-Var Coord (4B)	100.0%	52.6%	0.0%	0.0%

D. Reactive Power Distribution – Scenario 4 [Laptop]

--- D1. Volt-Var Local (4A) – per-DER Q statistics ---

98 DERs, 35136 timesteps with Q data

DER	mean Q	std Q	min Q	max Q	% at limit (>0.95*max)
75	-0.0045	0.0006	-0.0054	-0.0018	13.5%
13	-0.0042	0.0005	-0.0051	-0.0019	12.9%
74	-0.0047	0.0006	-0.0056	-0.0019	12.9%
29	-0.0044	0.0006	-0.0053	-0.0017	12.9%
12	-0.0045	0.0006	-0.0054	-0.0020	12.8%
28	-0.0044	0.0006	-0.0053	-0.0017	12.8%
11	-0.0041	0.0005	-0.0049	-0.0018	12.8%
73	-0.0042	0.0005	-0.0051	-0.0018	12.8%
27	-0.0043	0.0006	-0.0053	-0.0017	12.8%
72	-0.0040	0.0005	-0.0049	-0.0018	12.7%

Overall Q saturation: mean q_saturated_count = 0.0 of 98 DERs per timestep
Total reactive energy (from summary): 15663.66509132867 MVar.h

--- D2. Volt-Var Local (4A) – Q budget utilisation over time ---

Total |Q| per timestep: mean=1.783 MVar std=0.293 max=2.727 min=0.792

Monthly mean total |Q| (MVar):

Month 1: 1.611
Month 2: 1.643
Month 3: 1.736
Month 4: 1.843
Month 5: 1.881
Month 6: 1.770
Month 7: 1.869
Month 8: 1.841
Month 9: 1.831
Month 10: 1.813
Month 11: 1.656
Month 12: 1.895

--- D1. Volt-Var Coord (4B) – per-DER Q statistics ---

98 DERs, 35136 timesteps with Q data

DER	mean Q	std Q	min Q	max Q	% at limit (>0.95*max)
42	-0.1955	0.0130	-0.1966	-0.0226	99.0%
41	-0.1954	0.0134	-0.1966	0.0419	98.9%
40	-0.1923	0.0340	-0.1966	0.1966	98.2%
61	-0.0703	0.0119	-0.0723	0.0532	96.8%
60	-0.0703	0.0119	-0.0723	0.0532	96.8%
56	-0.0703	0.0119	-0.0723	0.0534	96.8%
55	-0.0702	0.0120	-0.0723	0.0723	96.8%
51	-0.0687	0.0183	-0.0723	0.0723	96.0%
50	-0.0682	0.0201	-0.0723	0.0723	95.9%
48	-0.0666	0.0245	-0.0723	0.0723	95.3%

Overall Q saturation: mean q_saturated_count = 33.6 of 98 DERs per timestep
Total reactive energy (from summary): 72927.28306952649 MVar.h

--- D2. Volt-Var Coord (4B) – Q budget utilisation over time ---

Total |Q| per timestep: mean=8.302 MVar std=2.546 max=16.292 min=0.792
 Monthly mean total |Q| (MVar):
 Month 1: 8.051
 Month 2: 8.091
 Month 3: 8.187
 Month 4: 8.319
 Month 5: 8.697
 Month 6: 7.705
 Month 7: 8.166
 Month 8: 8.016
 Month 9: 8.187
 Month 10: 8.411
 Month 11: 7.637
 Month 12: 10.105

E. Active Power Curtailment – Scenario 4 [Laptop]

--- E1. Volt-Var Local (4A) – curtailment overview ---

Curtailment steps: 35136
 Curtailed energy: 65138.2979 MWh
 Curtailment as % of total DER generation: 100.000%

Per-DER curtailment breakdown (top 10 most curtailed):

All timesteps are curtailment steps – cannot derive uncurtailed reference from non-curtailment steps.

Curtailment steps by month:

Month 1: 2976 steps
 Month 2: 2784 steps
 Month 3: 2976 steps
 Month 4: 2880 steps
 Month 5: 2976 steps
 Month 6: 2880 steps
 Month 7: 2976 steps
 Month 8: 2976 steps
 Month 9: 2880 steps
 Month 10: 2976 steps
 Month 11: 2880 steps
 Month 12: 2976 steps

--- E1. Volt-Var Coord (4B) – curtailment overview ---

Curtailment steps: 27851
 Curtailed energy: 52960.9922 MWh
 Curtailment as % of total DER generation: 81.305%

Per-DER curtailment breakdown (top 10 most curtailed):

DER	natural P (MW)	mean P on curt steps	curtailed MWh	curt steps
93	1.5003	0.1089	10054.5397	26932
91	1.2248	0.0889	8207.7875	26932
97	0.6124	0.0444	4103.8937	26932
0	0.4855	0.0365	3255.3674	27018
96	0.4369	0.0329	2929.8307	27018
92	0.4126	0.0311	2767.0623	27018
1	0.0000	0.0003	0.0000	0
2	0.0000	0.0003	0.0000	0
3	0.0000	0.0003	0.0000	0
4	0.0000	0.0003	0.0000	0

Sum of per-DER curtailed energy (curtailment steps only): 31318.4813 MWh

Summary curtailed_energy_mwh: 52960.9922 MWh (includes PT1-dynamics P lag on all steps, not just curtailment steps –

Curtailment steps by month:

Month 1: 1682 steps
 Month 2: 1735 steps
 Month 3: 2275 steps
 Month 4: 2649 steps
 Month 5: 2596 steps
 Month 6: 2701 steps
 Month 7: 2804 steps
 Month 8: 2755 steps
 Month 9: 2454 steps
 Month 10: 2310 steps
 Month 11: 2038 steps
 Month 12: 1852 steps

F. Export / Import Balance [Laptop]

Note: grid_import_mw < 0 means the network is NET EXPORTING (DER generation exceeds local load + losses).
 Scenario Export steps Import steps Export % Net export MWh Net import MWh

Volt-Var Local (4A)	35136	0	100.0%	95785.50	0.00
Volt-Var Coord (4B)	35136	0	100.0%	107057.95	0.00

--- F2. Monthly net export comparison: baseline vs Scenario 4 ---

Month	Volt-Var Local (4A)	Volt-Var Coord (4B)
1	7447.36	9161.43
2	6967.51	8312.61
3	7832.55	8662.44
4	8111.15	8424.23
5	8450.08	8925.79
6	8226.80	8369.67
7	8554.66	8822.65
8	8591.66	8829.22
9	8081.94	8672.78
10	8300.63	9429.98
11	7653.74	8637.47
12	7567.42	10809.69

(values in MWh exported per month)

G. Violation Frequency Map [Laptop]

--- G. Volt-Var Local (4A) ---

Total overvoltage bus-steps: 875065 | Total undervoltage bus-steps: 0

Top 10 most frequently overvoltage buses:

Bus	Count	% of all OV steps
44	35136	4.0%
45	35136	4.0%
46	35136	4.0%
47	35136	4.0%
43	35133	4.0%
42	35127	4.0%
68	33907	3.9%
67	33884	3.9%
66	33857	3.9%
98	33827	3.9%

No undervoltage bus-steps recorded.

--- G. Volt-Var Coord (4B) ---

Total overvoltage bus-steps: 477988 | Total undervoltage bus-steps: 0

Top 10 most frequently overvoltage buses:

Bus	Count	% of all OV steps
47	26632	5.6%
46	26611	5.6%
45	26512	5.5%
44	25917	5.4%
43	23246	4.9%
42	22337	4.7%
68	16639	3.5%
67	16614	3.5%
66	16573	3.5%
98	16503	3.5%

No undervoltage bus-steps recorded.

H. Summary Comparison Table [Laptop]

Metric	Volt-Var Local (4A)	Volt-Var Coord (4B)
Violation steps	35136	26632
OV bus-steps	875065	477988
UV bus-steps	0	0
VDI	114278.2644	89455.1786
Total losses (MWh)	1850.4283	2755.2821
Reactive energy (MVar·h)	15663.6651	72927.2831
Curtailed energy (MWh)	65138.2979	52960.9922
DER generation (MWh)	65138.2979	65138.2979
Grid export (MWh)	95785.5020	107057.9539
Peak max_vm_pu	1.0625	1.0644
Worst min_vm_pu	1.0135	0.9826
Peak line loading (%)	39.9026	84.9945
Coordination rate	0.0000	0.9948
Q saturation rate	0.0000	0.9948
Elapsed (s)	25122.0776	18072.5126

--- H2. Loss penalty and reactive cost vs baseline ---

END OF REPORT

Q_initial clip diagnostic (H1/H2)

```
=====
Q_INITIAL CLIP DIAGNOSTIC - scenario: volt_var_coord
=====
```

```
RPi records:      35136
Laptop records: 35136
Common timesteps: 35136
```

```
-----
PART A - q_saturated_count (out of up to 98 DERs), RPi vs Laptop
-----
```

```
RPi      : mean=33.62  median=30.0  max=92  min=0
Laptop   : mean=33.62  median=30.0  max=92  min=0
```

```
Per-timestep comparison (n=35136):
  RPi higher      : 97 (0.3%)
  Laptop higher   : 117 (0.3%)
  Equal           : 34922 (99.4%)
  Mean diff (RPi - Laptop): -0.000 DERs
```

```
>> Saturation counts are in the same order of magnitude.
>> Consistent with H1 (truncation pushing values over the
clip boundary only marginally more often on RPi).
```

```
-----
PART B - per-DER |q_mvar| magnitude comparison (q_mvar_by_sgen)
-----
```

```
Sampling 2067 of 35136 common timesteps (every 17th)
```

```
|q_rpi - q_laptop| across 202566 (DER, t) pairs:
  mean=0.000066  median=0.000012  max=0.393288 (MVar)
```

```
|q_rpi| / |q_laptop| ratio (excluding near-zero laptop values):
  mean=0.9999  median=1.0000
  fraction of ratios within +-10% of 1.0: 99.7%
```

```
>> The vast majority of (DER, t) pairs agree to within +-10%.
>> Consistent with H1 (truncation only bites at the margin
for a small subset of DERs/timesteps).
```

```
-----
PART C - top 15 largest individual |q_rpi - q_laptop| disagreements
-----
```

t= 28322	DER= 38	q_rpi=-0.19664	q_laptop=+0.19664	diff=0.39329	[sat_count RPi=41 Laptop=42]
t= 14008	DER= 38	q_rpi=-0.19664	q_laptop=+0.19664	diff=0.39329	[sat_count RPi=30 Laptop=29]
t= 9112	DER= 38	q_rpi=-0.19664	q_laptop=+0.19664	diff=0.39329	[sat_count RPi=30 Laptop=29]
t= 28322	DER= 93	q_rpi=+0.28617	q_laptop=+0.40740	diff=0.12123	[sat_count RPi=41 Laptop=42]
t= 9112	DER= 96	q_rpi=-0.46523	q_laptop=-0.36118	diff=0.10406	[sat_count RPi=30 Laptop=29]
t= 14008	DER= 96	q_rpi=-0.45237	q_laptop=-0.35229	diff=0.10008	[sat_count RPi=30 Laptop=29]
t= 25823	DER= 96	q_rpi=-0.60623	q_laptop=-0.69858	diff=0.09235	[sat_count RPi=30 Laptop=30]
t= 25398	DER= 96	q_rpi=-0.30556	q_laptop=-0.37845	diff=0.07289	[sat_count RPi=36 Laptop=37]
t= 9112	DER= 91	q_rpi=-0.11180	q_laptop=-0.15425	diff=0.04246	[sat_count RPi=30 Laptop=29]
t= 9112	DER= 87	q_rpi=+0.01605	q_laptop=-0.02626	diff=0.04231	[sat_count RPi=30 Laptop=29]
t= 9112	DER= 70	q_rpi=+0.00903	q_laptop=-0.03328	diff=0.04231	[sat_count RPi=30 Laptop=29]
t= 9112	DER= 85	q_rpi=+0.00515	q_laptop=-0.03716	diff=0.04230	[sat_count RPi=30 Laptop=29]
t= 9112	DER= 71	q_rpi=+0.00873	q_laptop=-0.03357	diff=0.04230	[sat_count RPi=30 Laptop=29]
t= 9112	DER= 82	q_rpi=+0.01302	q_laptop=-0.02928	diff=0.04230	[sat_count RPi=30 Laptop=29]
t= 9112	DER= 84	q_rpi=+0.00814	q_laptop=-0.03416	diff=0.04230	[sat_count RPi=30 Laptop=29]

```
=====
END OF REPORT
=====
```

Coordination ground-truth check

```
=====
COORDINATION GROUND-TRUTH CHECK - scenario: volt_var_coord
Thresholds: V_BAND=[0.95, 1.05] MIN_VIOLATION_DV=0.001
=====
```

```
-----
RPI - ground-truth check on coordination_active timesteps
-----
```

Timesteps with coordination_active=True: 34954 of 35136

Of 34954 active timesteps with usable vm_pu_by_bus data:

Real violation remained (margin < 0) : 18379 (52.6%)
Near the trigger's own noise floor (0 <= margin < 0.001) : 946 (2.7%)
Comfortably inside the band (margin >= 0.001) : 15629 (44.7%)

Margin distribution (pu, worst bus per timestep):

mean=0.00309 median=-0.00063 min=-0.01443 max=0.02500

>> 18379 timestep(s) show a REAL violation persisting
even after coordination. Dumping details below to check
correlation with curtailment_needed / curtail_exhausted /
q_saturated_count (i.e. was Q headroom already maxed out).

Full detail on the 18379 real-violation timestep(s):

```
t=0 timestamp=2016-01-01T00:00:00+01:00 margin=-0.00068
  q_saturated_count      = 36
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0506759733523017 / 1.009191447343097
  over_voltage_buses     = [42, 43, 44, 45, 46, 47]
  under_voltage_buses    = []
t=1 timestamp=2016-01-01T00:15:00+01:00 margin=-0.00613
  q_saturated_count      = 35
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0561318644901034 / 1.0154910820993313
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 97]
  under_voltage_buses    = []
t=7 timestamp=2016-01-01T01:45:00+01:00 margin=-0.00003
  q_saturated_count      = 36
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0500287716702532 / 1.009487838001714
  over_voltage_buses     = [42, 43, 44, 45, 46, 47]
  under_voltage_buses    = []
t=8 timestamp=2016-01-01T02:00:00+01:00 margin=-0.00658
  q_saturated_count      = 35
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0565751188080141 / 1.0161418750659161
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 97]
  under_voltage_buses    = []
t=13 timestamp=2016-01-01T03:15:00+01:00 margin=-0.00101
  q_saturated_count      = 36
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0510143041667959 / 1.010254883627266
  over_voltage_buses     = [42, 43, 44, 45, 46, 47]
  under_voltage_buses    = []
t=14 timestamp=2016-01-01T03:30:00+01:00 margin=-0.00606
  q_saturated_count      = 38
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0560561716612618 / 1.0173004266540833
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 97]
  under_voltage_buses    = []
t=15 timestamp=2016-01-01T03:45:00+01:00 margin=-0.00617
  q_saturated_count      = 38
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0561656420276322 / 1.0171713708974544
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 97]
  under_voltage_buses    = []
t=16 timestamp=2016-01-01T04:00:00+01:00 margin=-0.00622
  q_saturated_count      = 37
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0562242346890274 / 1.0169171249356574
```

```
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95]
    under_voltage_buses = []
t=17 timestamp=2016-01-01T04:15:00+01:00 margin=-0.00609
    q_saturated_count = 37
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0560911952964838 / 1.0170178606152962
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95]
    under_voltage_buses = []
t=18 timestamp=2016-01-01T04:30:00+01:00 margin=-0.00649
    q_saturated_count = 37
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0564860990481504 / 1.0172104832413997
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95]
    under_voltage_buses = []
t=19 timestamp=2016-01-01T04:45:00+01:00 margin=-0.00846
    q_saturated_count = 36
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.058457161584332 / 1.017327296050351
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95]
    under_voltage_buses = []
t=20 timestamp=2016-01-01T05:00:00+01:00 margin=-0.00693
    q_saturated_count = 37
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0569347210775062 / 1.0167516250376232
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95]
    under_voltage_buses = []
t=21 timestamp=2016-01-01T05:15:00+01:00 margin=-0.00629
    q_saturated_count = 36
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0562900539802502 / 1.0161711906749282
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95]
    under_voltage_buses = []
t=23 timestamp=2016-01-01T05:45:00+01:00 margin=-0.00023
    q_saturated_count = 36
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0502329637877625 / 1.0088619720997274
    over_voltage_buses = [42, 43, 44, 45, 46, 47]
    under_voltage_buses = []
t=24 timestamp=2016-01-01T06:00:00+01:00 margin=-0.00106
    q_saturated_count = 36
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0510644720742957 / 1.0088723926995948
    over_voltage_buses = [42, 43, 44, 45, 46, 47]
    under_voltage_buses = []
t=25 timestamp=2016-01-01T06:15:00+01:00 margin=-0.00157
    q_saturated_count = 36
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.051566445766697 / 1.009283503991686
    over_voltage_buses = [42, 43, 44, 45, 46, 47]
    under_voltage_buses = []
t=26 timestamp=2016-01-01T06:30:00+01:00 margin=-0.00871
    q_saturated_count = 36
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.058714499786784 / 1.0167508442115016
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95]
    under_voltage_buses = []
t=27 timestamp=2016-01-01T06:45:00+01:00 margin=-0.00753
    q_saturated_count = 35
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0575256275396538 / 1.0166877291422012
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95]
    under_voltage_buses = []
t=28 timestamp=2016-01-01T07:00:00+01:00 margin=-0.00694
    q_saturated_count = 35
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0569447353514854 / 1.0168977695771508
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95]
    under_voltage_buses = []
t=34 timestamp=2016-01-01T08:30:00+01:00 margin=-0.00631
    q_saturated_count = 36
```

```
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0563096116987136 / 1.0163390857976504
over_voltage_buses = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 94, 95, 96, 98]
under_voltage_buses = []
t=98 timestamp=2016-01-02T00:30:00+01:00 margin=-0.00129
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.051290307748084 / 1.0194416659900452
over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=99 timestamp=2016-01-02T00:45:00+01:00 margin=-0.00061
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0506142564229155 / 1.0183238383091804
over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=100 timestamp=2016-01-02T01:00:00+01:00 margin=-0.00095
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0509491330028389 / 1.0184741327774063
over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=101 timestamp=2016-01-02T01:15:00+01:00 margin=-0.00068
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.050675334743881 / 1.018110222152425
over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=102 timestamp=2016-01-02T01:30:00+01:00 margin=-0.00096
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.05096403178839 / 1.0185339984368469
over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=103 timestamp=2016-01-02T01:45:00+01:00 margin=-0.00155
q_saturated_count = 31
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0515535821852118 / 1.0188218344978066
over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=104 timestamp=2016-01-02T02:00:00+01:00 margin=-0.00114
q_saturated_count = 31
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0511400547875824 / 1.0184563471265928
over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=109 timestamp=2016-01-02T03:15:00+01:00 margin=-0.00108
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0510761027031443 / 1.0197862901638255
over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=110 timestamp=2016-01-02T03:30:00+01:00 margin=-0.00147
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0514732116232821 / 1.0194689480722559
over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=111 timestamp=2016-01-02T03:45:00+01:00 margin=-0.00169
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.05169140201154 / 1.0194700180548368
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=112 timestamp=2016-01-02T04:00:00+01:00 margin=-0.00238
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0523759551529093 / 1.0198969686300443
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
```

```

under_voltage_buses = []
t=113 timestamp=2016-01-02T04:15:00+01:00 margin=-0.00243
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.052432683475007 / 1.0197539558425681
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=114 timestamp=2016-01-02T04:30:00+01:00 margin=-0.00253
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.052532094662751 / 1.0200698274481177
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=115 timestamp=2016-01-02T04:45:00+01:00 margin=-0.00291
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0529145274900813 / 1.0203053179473538
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=116 timestamp=2016-01-02T05:00:00+01:00 margin=-0.00179
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.05178755458085 / 1.019376287056505
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=117 timestamp=2016-01-02T05:15:00+01:00 margin=-0.00006
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.050062482674225 / 1.0187216677841227
over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=144 timestamp=2016-01-02T12:00:00+01:00 margin=-0.00105
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0510500524826447 / 1.021398176089376
over_voltage_buses = [42, 43, 44, 45, 46, 47, 68, 91, 92, 93, 94, 95, 96]
under_voltage_buses = []
t=153 timestamp=2016-01-02T14:15:00+01:00 margin=-0.00176
q_saturated_count = 28
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0517574527246916 / 1.0227792692975464
over_voltage_buses = [42, 43, 44, 45, 46, 47, 91, 92, 93, 94, 95, 96]
under_voltage_buses = []
t=154 timestamp=2016-01-02T14:30:00+01:00 margin=-0.00268
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0526786070716103 / 1.0215524951507122
over_voltage_buses = [42, 43, 44, 45, 46, 47, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=155 timestamp=2016-01-02T14:45:00+01:00 margin=-0.00119
q_saturated_count = 29
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0511852062845202 / 1.0206547182133745
over_voltage_buses = [42, 43, 44, 45, 46, 47, 91, 92, 93, 94, 95, 96]
under_voltage_buses = []
t=191 timestamp=2016-01-02T23:45:00+01:00 margin=-0.00242
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0524243513319866 / 1.0210460743282586
over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=192 timestamp=2016-01-03T00:00:00+01:00 margin=-0.00260
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0525970039820793 / 1.021159236424415
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=199 timestamp=2016-01-03T01:45:00+01:00 margin=-0.00063
q_saturated_count = 30
curtailment_needed = True

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curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.050633499719611 / 1.0204571682876096
over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=200 timestamp=2016-01-03T02:00:00+01:00 margin=-0.00265
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0526464853501416 / 1.0211453347160588
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=201 timestamp=2016-01-03T02:15:00+01:00 margin=-0.00267
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0526748054541395 / 1.0211553015666488
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=202 timestamp=2016-01-03T02:30:00+01:00 margin=-0.00271
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.052711420086715 / 1.0211472792988785
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=203 timestamp=2016-01-03T02:45:00+01:00 margin=-0.00291
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.052908114308148 / 1.021241510731518
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=204 timestamp=2016-01-03T03:00:00+01:00 margin=-0.00353
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0535317280131036 / 1.0215612869956308
over_voltage_buses = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=205 timestamp=2016-01-03T03:15:00+01:00 margin=-0.00214
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.052140615980558 / 1.0210828504019602
over_voltage_buses = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=206 timestamp=2016-01-03T03:30:00+01:00 margin=-0.00087
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0508714699059727 / 1.0207081464784358
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=207 timestamp=2016-01-03T03:45:00+01:00 margin=-0.00080
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.050795452165205 / 1.0206927341795586
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=208 timestamp=2016-01-03T04:00:00+01:00 margin=-0.00008
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0500793655283098 / 1.020383227812664
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=209 timestamp=2016-01-03T04:15:00+01:00 margin=-0.00133
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0513349606261098 / 1.0208034380167017
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
t=210 timestamp=2016-01-03T04:30:00+01:00 margin=-0.00450
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0545025506223349 / 1.022753048781504
over_voltage_buses = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 90, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses = []
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t=211 timestamp=2016-01-03T04:45:00+01:00 margin=-0.00302
  q_saturated_count      = 30
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.053021052271288 / 1.0213246784433354
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
  under_voltage_buses    = []
t=212 timestamp=2016-01-03T05:00:00+01:00 margin=-0.00261
  q_saturated_count      = 30
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0526141925098385 / 1.021133604853293
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
  under_voltage_buses    = []
t=213 timestamp=2016-01-03T05:15:00+01:00 margin=-0.00319
  q_saturated_count      = 30
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0531927803551406 / 1.021409937479619
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
  under_voltage_buses    = []
t=214 timestamp=2016-01-03T05:30:00+01:00 margin=-0.00430
  q_saturated_count      = 30
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0542959636786575 / 1.0226479254970584
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 90, 91, 92, 93, 94, 95, 96, 98]
  under_voltage_buses    = []
t=215 timestamp=2016-01-03T05:45:00+01:00 margin=-0.00115
  q_saturated_count      = 30
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0511526386093968 / 1.0204464553122419
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
  under_voltage_buses    = []
t=227 timestamp=2016-01-03T08:45:00+01:00 margin=-0.00009
  q_saturated_count      = 30
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0500909825961107 / 1.020705380994442
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 94, 95, 96, 98]
  under_voltage_buses    = []
t=228 timestamp=2016-01-03T09:00:00+01:00 margin=-0.00090
  q_saturated_count      = 30
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0509009315260531 / 1.0221856552126158
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
  under_voltage_buses    = []
t=290 timestamp=2016-01-04T00:30:00+01:00 margin=-0.00181
  q_saturated_count      = 30
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0518126185871457 / 1.020426719856361
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
  under_voltage_buses    = []
t=291 timestamp=2016-01-04T00:45:00+01:00 margin=-0.00278
  q_saturated_count      = 30
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0527802773018375 / 1.0208512218012757
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
  under_voltage_buses    = []
t=292 timestamp=2016-01-04T01:00:00+01:00 margin=-0.00325
  q_saturated_count      = 30
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0532495797254495 / 1.0210381554724883
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
  under_voltage_buses    = []
t=293 timestamp=2016-01-04T01:15:00+01:00 margin=-0.00349
  q_saturated_count      = 30
  curtailment_needed     = True
  curtail_exhausted      = True
  max_vm_pu / min_vm_pu  = 1.0534899743165684 / 1.0212366311709349
  over_voltage_buses     = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
  under_voltage_buses    = []
t=294 timestamp=2016-01-04T01:30:00+01:00 margin=-0.00348
  q_saturated_count      = 30
  curtailment_needed     = True
  curtail_exhausted      = True
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max_vm_pu / min_vm_pu = 1.0534821791194826 / 1.0212611791914687
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
under_voltage_buses = []]
t=295 timestamp=2016-01-04T01:45:00+01:00 margin=-0.00204
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0520444777856959 / 1.0206505329179347
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
under_voltage_buses = []]
t=296 timestamp=2016-01-04T02:00:00+01:00 margin=-0.00079
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0507895028633127 / 1.0201992643772453
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
under_voltage_buses = []]
t=299 timestamp=2016-01-04T02:45:00+01:00 margin=-0.00143
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.051429865633835 / 1.0202281447670454
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
under_voltage_buses = []]
t=300 timestamp=2016-01-04T03:00:00+01:00 margin=-0.00247
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0524724667995593 / 1.0205511440811905
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
under_voltage_buses = []]
t=301 timestamp=2016-01-04T03:15:00+01:00 margin=-0.00214
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0521436683743215 / 1.0198337783352582
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
under_voltage_buses = []]
t=302 timestamp=2016-01-04T03:30:00+01:00 margin=-0.00253
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0525327157964581 / 1.0198483010525454
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
under_voltage_buses = []]
t=303 timestamp=2016-01-04T03:45:00+01:00 margin=-0.00241
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0524115528497844 / 1.0198191182366858
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
under_voltage_buses = []]
t=304 timestamp=2016-01-04T04:00:00+01:00 margin=-0.00263
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0526342071122234 / 1.0198695814521583
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
under_voltage_buses = []]
t=305 timestamp=2016-01-04T04:15:00+01:00 margin=-0.00210
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.05209553682742 / 1.0205170816967992
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
under_voltage_buses = []]
t=309 timestamp=2016-01-04T05:15:00+01:00 margin=-0.00222
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0522200230950174 / 1.0211282996602085
over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
under_voltage_buses = []]
t=310 timestamp=2016-01-04T05:30:00+01:00 margin=-0.00317
q_saturated_count = 30
curtailment_needed = True
curtail_exhausted = True
max_vm_pu / min_vm_pu = 1.0531701717859752 / 1.0214481490454224
over_voltage_buses = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95,
under_voltage_buses = []]
t=311 timestamp=2016-01-04T05:45:00+01:00 margin=-0.00294
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q_saturated_count      = 30
curtailment_needed     = True
curtail_exhausted      = True
max_vm_pu / min_vm_pu  = 1.0529390834937635 / 1.0212433499976588
over_voltage_buses     = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
under_voltage_buses     = []
t=312 timestamp=2016-01-04T06:00:00+01:00 margin=-0.00254
q_saturated_count      = 30
curtailment_needed     = True
curtail_exhausted      = True
max_vm_pu / min_vm_pu  = 1.0525446815417587 / 1.0209381484267435
over_voltage_buses     = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
under_voltage_buses     = []
t=313 timestamp=2016-01-04T06:15:00+01:00 margin=-0.00103
q_saturated_count      = 30
curtailment_needed     = True
curtail_exhausted      = True
max_vm_pu / min_vm_pu  = 1.0510342964729962 / 1.0192833390697564
over_voltage_buses     = [42, 43, 44, 45, 46, 47, 91, 92, 93, 94, 95, 96]
under_voltage_buses     = []
t=314 timestamp=2016-01-04T06:30:00+01:00 margin=-0.00106
q_saturated_count      = 30
curtailment_needed     = True
curtail_exhausted      = True
max_vm_pu / min_vm_pu  = 1.0510620797197965 / 1.0190905769950622
over_voltage_buses     = [42, 43, 44, 45, 46, 47, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96, 98]
under_voltage_buses     = []
t=323 timestamp=2016-01-04T08:45:00+01:00 margin=-0.00249
q_saturated_count      = 28
curtailment_needed     = True
curtail_exhausted      = True
max_vm_pu / min_vm_pu  = 1.0524884453471914 / 1.0218283384891529
over_voltage_buses     = [42, 43, 44, 45, 46, 47, 91, 92, 93, 94, 95, 96]
under_voltage_buses     = []
t=324 timestamp=2016-01-04T09:00:00+01:00 margin=-0.00515
q_saturated_count      = 7
curtailment_needed     = True
curtail_exhausted      = True
max_vm_pu / min_vm_pu  = 1.0551488480291373 / 1.0247264917497139
over_voltage_buses     = [42, 43, 44, 45, 46, 47, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91,
under_voltage_buses     = []
t=334 timestamp=2016-01-04T11:30:00+01:00 margin=-0.00160
q_saturated_count      = 30
curtailment_needed     = True
curtail_exhausted      = True
max_vm_pu / min_vm_pu  = 1.0515950788728317 / 1.01951944055721
over_voltage_buses     = [42, 43, 44, 45, 46, 47, 91, 92, 93, 94, 95, 96]
under_voltage_buses     = []
t=335 timestamp=2016-01-04T11:45:00+01:00 margin=-0.00146
q_saturated_count      = 30
curtailment_needed     = True
curtail_exhausted      = True
max_vm_pu / min_vm_pu  = 1.0514628361075393 / 1.019339340743218
over_voltage_buses     = [42, 43, 44, 45, 46, 47, 91, 92, 93, 94, 95, 96]
under_voltage_buses     = []
t=336 timestamp=2016-01-04T12:00:00+01:00 margin=-0.00166
q_saturated_count      = 30
curtailment_needed     = True
curtail_exhausted      = True
max_vm_pu / min_vm_pu  = 1.0516607221026064 / 1.0195004286381106
over_voltage_buses     = [42, 43, 44, 45, 46, 47, 91, 92, 93, 94, 95, 96]
under_voltage_buses     = []
t=337 timestamp=2016-01-04T12:15:00+01:00 margin=-0.00203
q_saturated_count      = 30
curtailment_needed     = True
curtail_exhausted      = True
max_vm_pu / min_vm_pu  = 1.0520273211746014 / 1.0196064452541294
over_voltage_buses     = [42, 43, 44, 45, 46, 47, 91, 92, 93, 94, 95, 96]
under_voltage_buses     = []
t=486 timestamp=2016-01-06T01:30:00+01:00 margin=-0.00695
q_saturated_count      = 37
curtailment_needed     = True
curtail_exhausted      = True
max_vm_pu / min_vm_pu  = 1.056954961148886 / 1.017053144981359
over_voltage_buses     = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95,
under_voltage_buses     = []
t=487 timestamp=2016-01-06T01:45:00+01:00 margin=-0.00846
q_saturated_count      = 36
curtailment_needed     = True
curtail_exhausted      = True
max_vm_pu / min_vm_pu  = 1.0584558522000609 / 1.017743287954375

```

```
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94,
    under_voltage_buses = []
t=488 timestamp=2016-01-06T02:00:00+01:00 margin=-0.00727
    q_saturated_count = 38
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0572709289452356 / 1.0185478472626677
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 90, 91,
    under_voltage_buses = []
t=489 timestamp=2016-01-06T02:15:00+01:00 margin=-0.00657
    q_saturated_count = 38
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0565714799994441 / 1.0176167315995952
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94,
    under_voltage_buses = []
t=490 timestamp=2016-01-06T02:30:00+01:00 margin=-0.00631
    q_saturated_count = 39
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0563096948519313 / 1.016915127537437
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94,
    under_voltage_buses = []
t=491 timestamp=2016-01-06T02:45:00+01:00 margin=-0.00696
    q_saturated_count = 36
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0569615631442775 / 1.0157821393858744
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95,
    under_voltage_buses = []
t=495 timestamp=2016-01-06T03:45:00+01:00 margin=-0.00453
    q_saturated_count = 37
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0545340965998855 / 1.0130679584040165
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 95, 96, 98]
    under_voltage_buses = []
t=496 timestamp=2016-01-06T04:00:00+01:00 margin=-0.00566
    q_saturated_count = 37
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.055656622411689 / 1.0132497894527008
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
    under_voltage_buses = []
t=497 timestamp=2016-01-06T04:15:00+01:00 margin=-0.00281
    q_saturated_count = 41
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0528142371482474 / 1.0116741484113123
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 91, 92, 93, 94, 95, 96,
    under_voltage_buses = []
t=498 timestamp=2016-01-06T04:30:00+01:00 margin=-0.00278
    q_saturated_count = 41
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0527801250955215 / 1.0104262678952456
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 92, 93, 94, 95, 96, 98]
    under_voltage_buses = []
t=499 timestamp=2016-01-06T04:45:00+01:00 margin=-0.00193
    q_saturated_count = 41
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0519315110891845 / 1.009201042157086
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 98]
    under_voltage_buses = []
t=500 timestamp=2016-01-06T05:00:00+01:00 margin=-0.00103
    q_saturated_count = 41
    curtailment_needed = True
    curtail_exhausted = True
    max_vm_pu / min_vm_pu = 1.0510274026797455 / 1.007958300854363
    over_voltage_buses = [42, 43, 44, 45, 46, 47, 60, 61, 62, 63, 64, 65, 66, 67, 68, 98]
    under_voltage_buses = []
t=526 timestamp=2016-01-06T11:30:00+01:00 margin=-0.00323
    q_saturated_count = 30
    curtailment_needed = True
    curtail_exhausted
```